National University of Computing and Emerging Sciences

Department of Computing

Compiler COnstruction

Assignment#01

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Submitted To:

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Section: F

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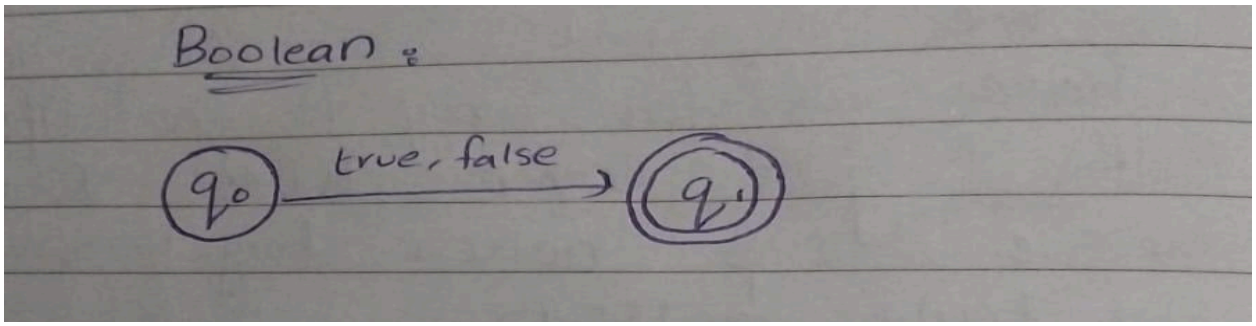
Automata Design

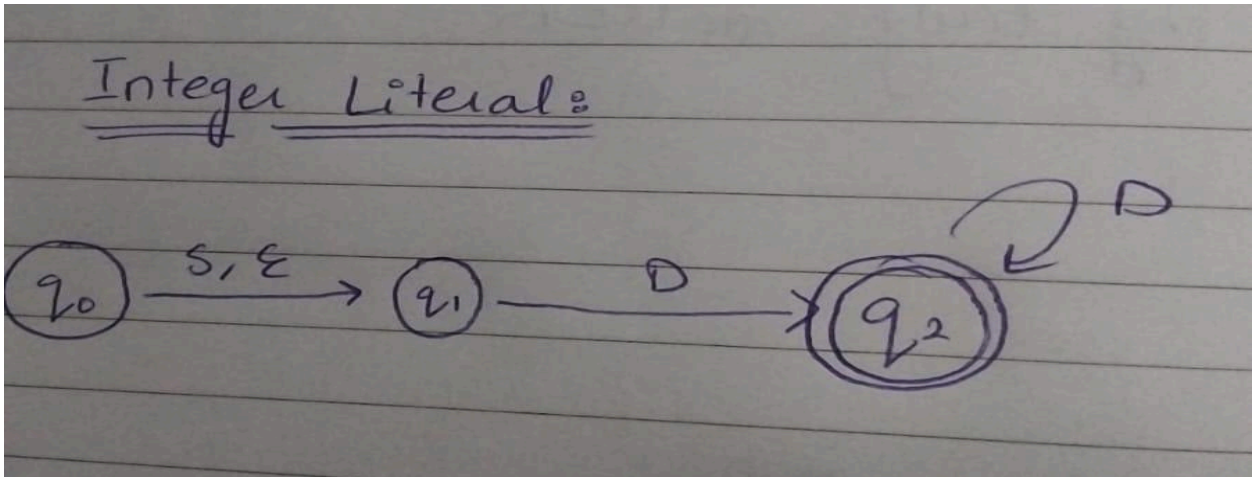
Regular Expressions:

1. Σ(L + N + U)* , Σ = [A-Z], L = [a-z], N = [0-9], U = [_]
2. (λ + S)NN*
3. (λ + S)(NN*DNN*(Σ(S + λ)NN*) + λ)
4. (true + false)
5. ww*
6. Single Line Comment: ##(cc* + λ)
7. “(c + \ (“ +\ + n + t + r))*”

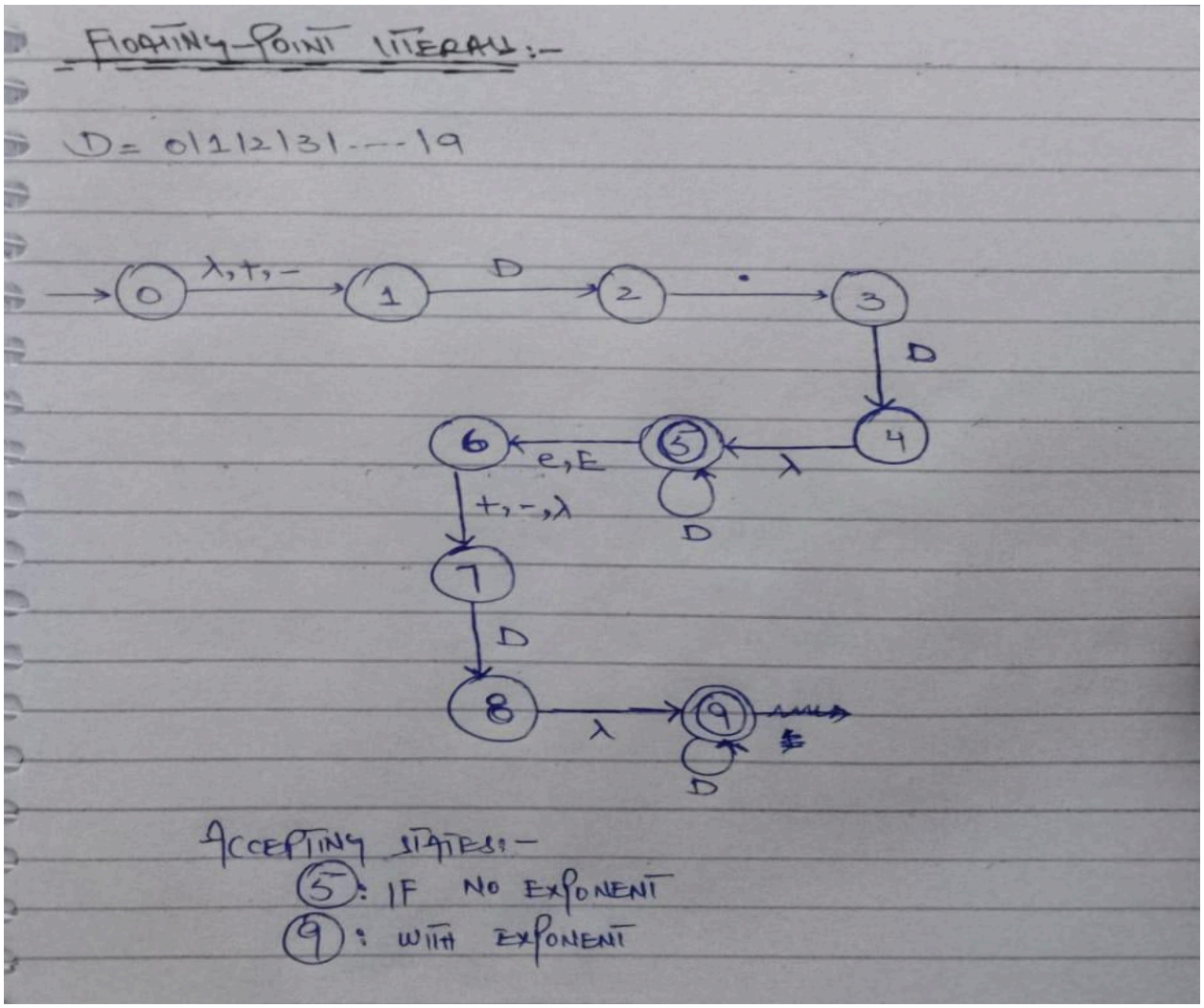
Boolean:



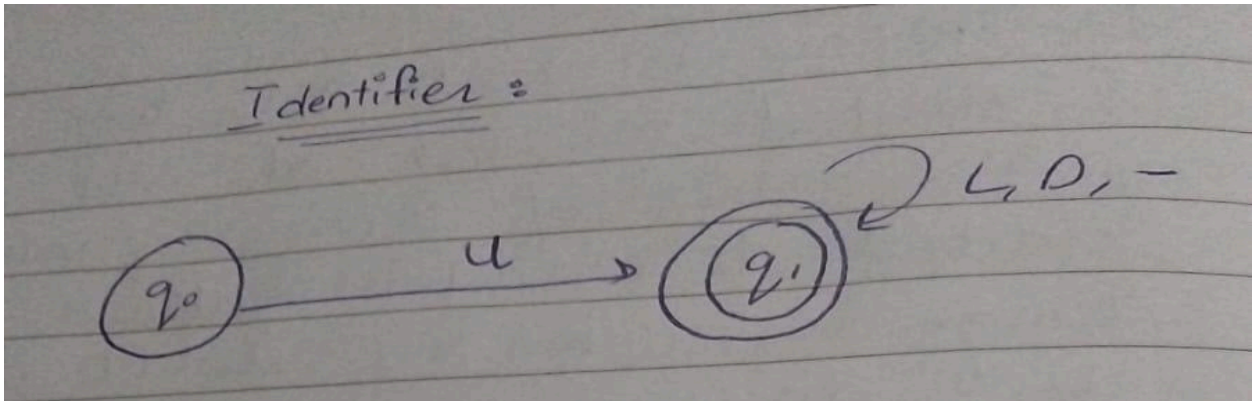
Integer Literal:



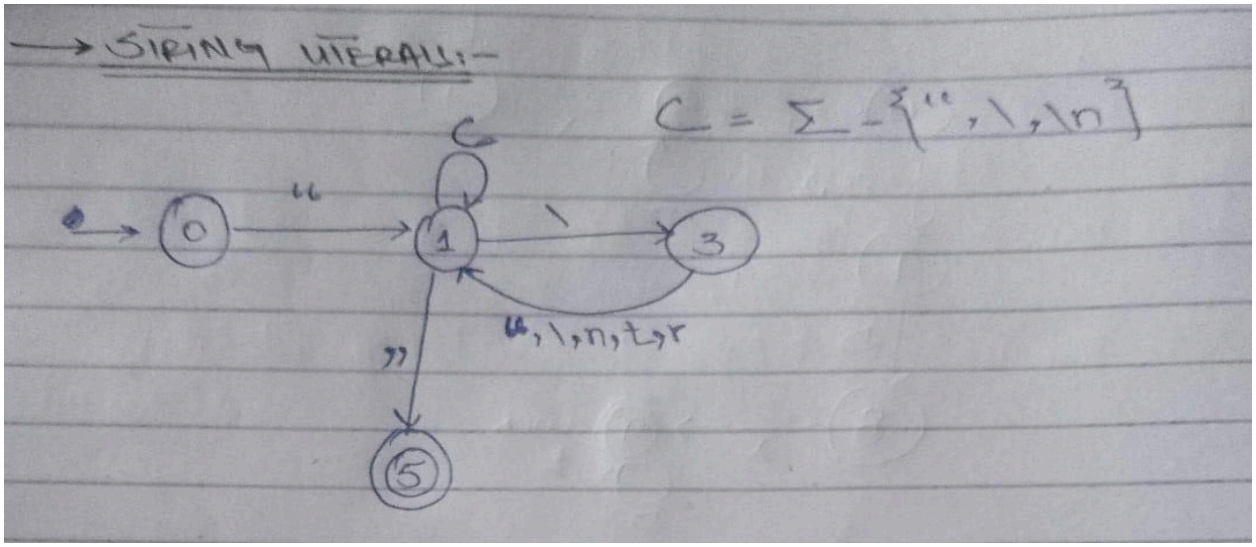
Floating-Point Literals:



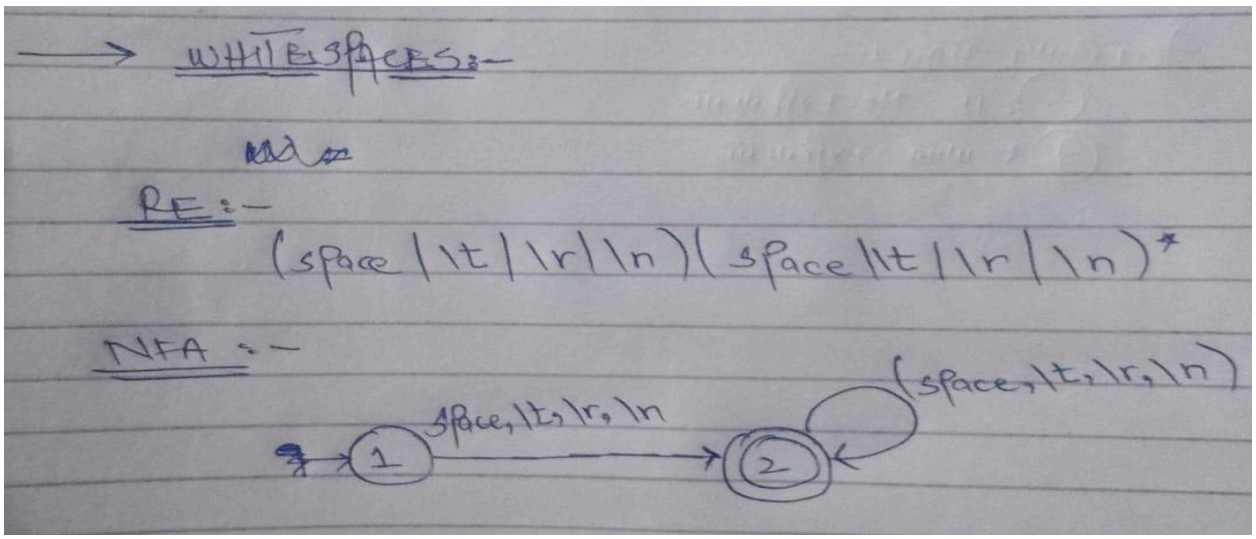
Identifiers:



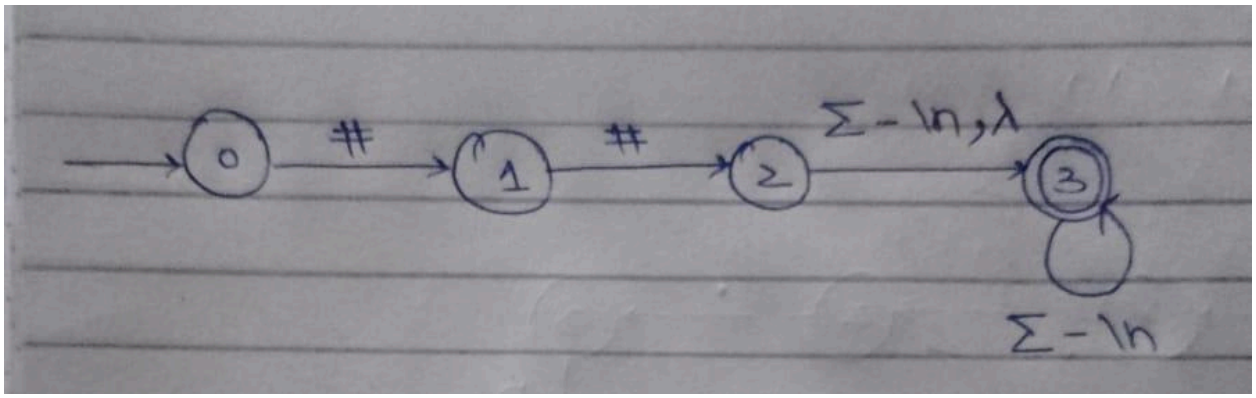
String:



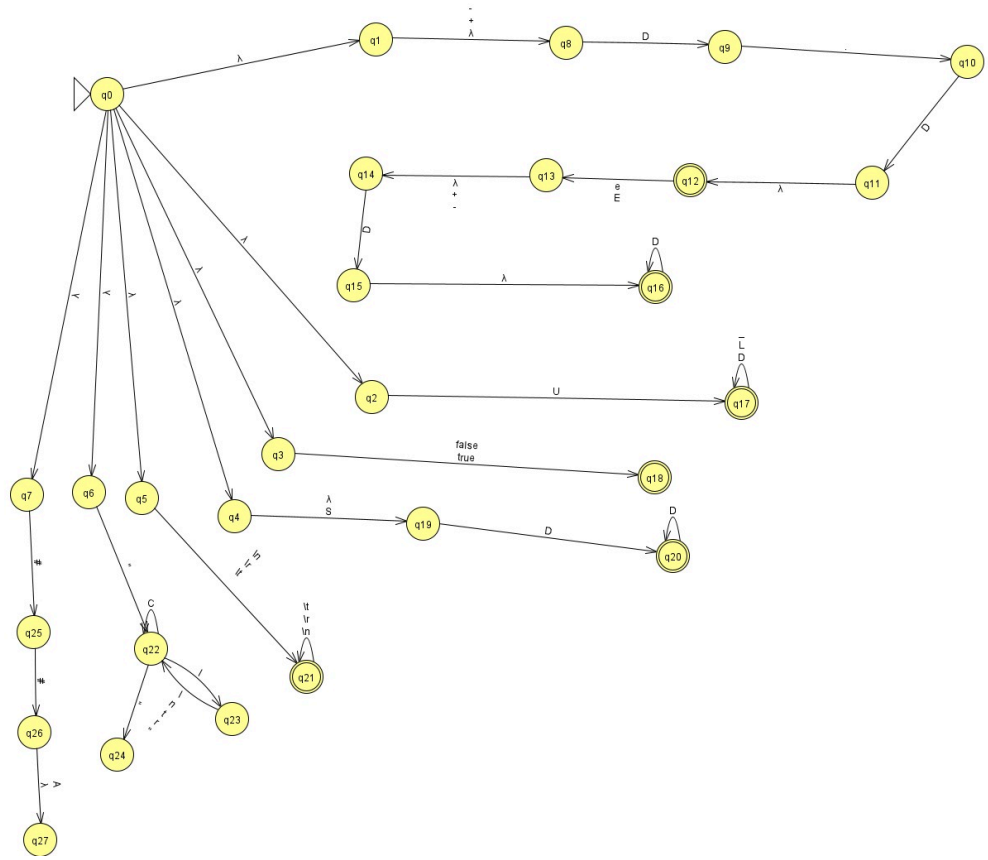
WhiteSpaces:



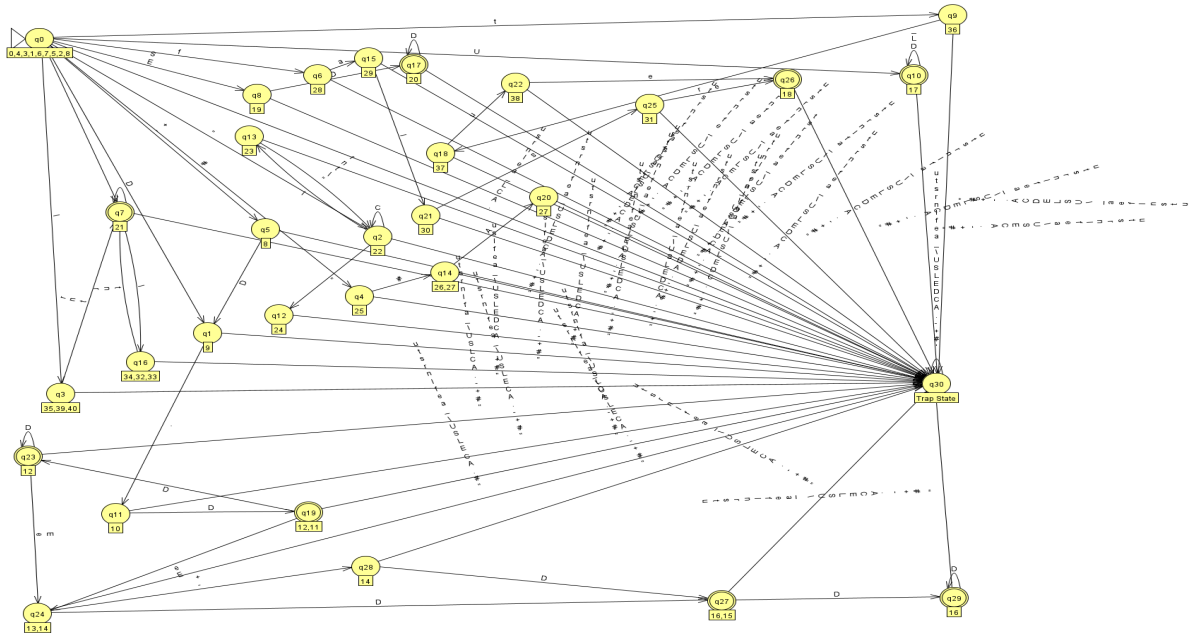
Single-Line Comments:



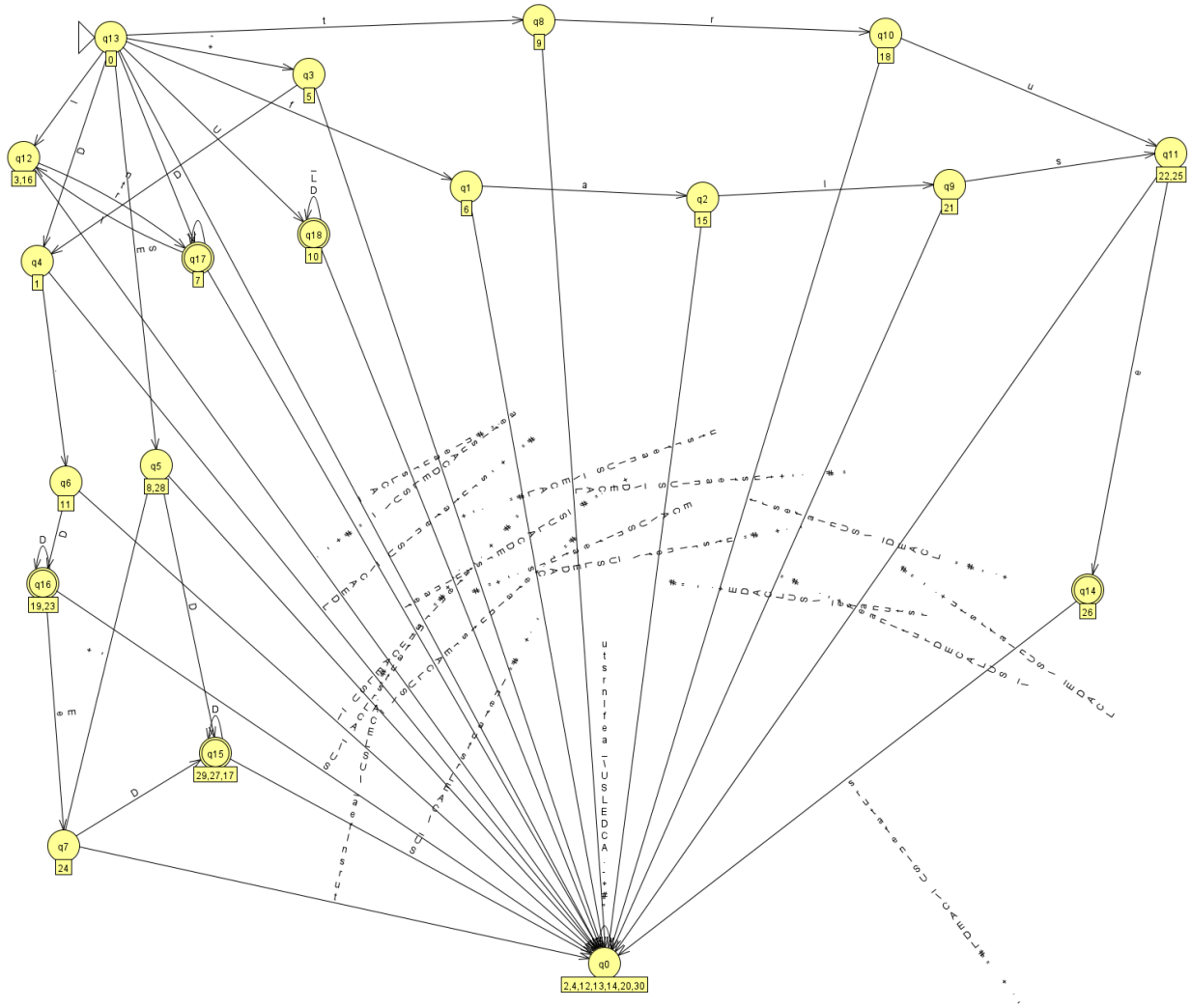
NFA:


q0
q1
q2
q3
q4
q5
q6
q7
q8
q9
q10
q11
q12
q13
q14
q15
q16
q17
q18
q19
q20
q21
q22
q23
q24
q25
q26
q27
false
true

DFA:


q0
0,4,3,1,6,7,5,2,8
q9
36
q10
17
q15
29
q17
20
q6
28
q22
38
q26
18
q8
19
q25
31
q13
23
q18
37
q7
21
q20
22
q21
30
q5
8
q2
22
q14
26,27
q4
25
q12
24
q1
9
q16
34,32,33
q3
35,39,40
q30
Trap State
q23
12
q11
10
q19
12,11
q28
14
q24
13,14
q27
16,15
q29
16

Minimized DFA:



Transition Table Minimized DFA:

Current State	Input Symbol	Next State	Token Type
q0	D (digit)	q2	INTEGER_LITERAL
q0	S (+ / -)	q7	—
q0	" (quote)	q5	STRING_LITERAL
q0	U (uppercase letter)	q9	IDENTIFIER
q0	# (hash)	q4	COMMENT
q0	\ (backslash)	q1	ESCAPE_SEQ
q0	t (letter)	q3	KEYWORD
q0	f (letter)	q8	BOOLEAN / KEYWORD
q0	space	q6	WHITESPACE
q1	n, r, t, "	q6	ESCAPE_VALID
q2	D (digit)	q2	INTEGER_LITERAL
q2	. (dot)	q12	FLOAT_LITERAL
q2	E, e	q20	FLOAT_LITERAL
q3	r	q13	—
q4	#	q14	COMMENT
q5	" (quote)	q16	STRING_LITERAL
q5	C (valid char)	q5	STRING_LITERAL
q5	\ (backslash)	q15	ESCAPE_SEQ
q6	space	q6	WHITESPACE
q6	\ (backslash)	q17	ESCAPE_VALID
q6	r, n, t	q6	ESCAPE_VALID
q7	D (digit)	q18	INTEGER_LITERAL

q8	a	q19	—
q9	D (digit)	q9	IDENTIFIER
q9	L (lowercase letter)	q9	IDENTIFIER
q9	_ (underscore)	q9	IDENTIFIER
q10	D (digit)	q11	INTEGER_LITERAL
q11	D (digit)	q11	INTEGER_LITERAL
q12	D (digit)	q20	FLOAT_LITERAL
q13	r	q21	—
q14	A (any character)	q22	COMMENT
q15	\	q5	STRING_LITERAL
q15	n	q5	STRING_LITERAL
q15	t	q5	STRING_LITERAL
q15	r	q5	STRING_LITERAL
q15	"	q5	STRING_LITERAL
q17	r, n, t, "	q6	ESCAPE_VALID
q18	D (digit)	q18	INTEGER_LITERAL
q19	l	q23	—
q20	D (digit)	q27	FLOAT_LITERAL
q20	E, e	q24	FLOAT_LITERAL
q21	u	q25	BOOLEAN_LITERAL
q23	s	q26	—
q24	+, -	q28	—
q24	D (digit)	q28	FLOAT_LITERAL
q26	e	q25	BOOLEAN_LITERAL

q27	D (digit)	q29	FLOAT_LITERAL
q28	D (digit)	q27	FLOAT_LITERAL
q29	D (digit)	q29	FLOAT_LITERAL